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Lab Report

# Anti-Phishing Defense – Netcraft Browser Extension

# Contents

1. Executive Summary ..... 3
2. Introduction to Netcraft ..... 4
2.1 Tool Overview ..... 4
2.2 Key Capabilities ..... 5
3. Extension Installation ..... 6
3.1 Locating the Extension — Netcraft Website ..... 6
3.2 Choosing a Browser ..... 6
3.3 Installing from the Firefox Add-Ons Store ..... 7
3.4 Confirming Installation ..... 7
4. Using the Netcraft Extension ..... 8
4.1 Live Site Report Panel ..... 8
4.2 Reporting a Malicious URL ..... 9
5. Full Netcraft Site Report ..... 9
5.1 Accessing the Site Report ..... 9
5.2 Background & Network Information ..... 10
5.3 IP Delegation ..... 10
5.4 IP Geolocation ..... 11
6. Netcraft Trends ..... 12
6.1 Viewing the Trends List ..... 12
7. Findings Summary ..... 13
8. Netcraft as a Defensive Countermeasure ..... 14
9. Conclusion ..... 14

# 1. Executive Summary

This report documents the installation and usage of the Netcraft Anti-Phishing Browser Extension as a defensive countermeasure against the types of phishing attacks simulated in the SET and Gophish lab exercises. Netcraft provides real-time protection against malicious websites, credential-harvesting pages, and suspicious URLs — directly within the browser.

The lab walkthrough covers:

- Locating and installing the Netcraft browser extension from the official Netcraft website and the Firefox Add-ons store.
- Verifying extension installation and confirming it is active in the browser toolbar.
- Using the extension's live Site Report panel to inspect a website's risk profile, hosting information, and site rank directly from the browser.
- Navigating to the full Netcraft Site Report page to review detailed network intelligence — including background information, hosting company, IP delegation, and geolocation data.
- Using the Netcraft Trends tool to explore site ranking data and hosting history.
- Reporting a malicious URL to Netcraft through the extension's built-in fraud reporting feature.

**Note:** Netcraft is used here from a defender's perspective — to demonstrate how security-aware users and organizations can detect and report the exact types of phishing pages built in the SET and Gophish labs. Understanding the attacker's tools and the defender's countermeasures together provides a complete picture of the phishing attack lifecycle.

# 2. Introduction to Netcraft

Netcraft is a UK-based internet security company founded in 1994. It is best known for its web server survey work and its anti-phishing services. The Netcraft browser extension provides free, real-time protection against phishing sites, malicious JavaScript, and fraudulent pages — and allows users to report suspicious URLs directly to Netcraft's takedown service.

From a penetration testing and security awareness perspective, Netcraft represents the kind of defensive tooling that attackers must consider: understanding how Netcraft detects and flags malicious pages helps red teamers understand what triggers detection, and helps defenders understand what protections are available.




## 2.1 Tool Overview

| | |
|---|---|
| **Tool Name** | Netcraft Browser Extension |
| **Extension Type** | Anti-phishing, site intelligence, fraud reporting |
| **Supported Browsers** | Chrome, Firefox, Edge, Opera |
| **License** | Free — available from browser extension stores |
| **Website** | https://www.netcraft.com |
| **Extension URL** | netcraft.com/resources/apps-and-extensions/browser-extension |
| **Site Reports** | https://sitereport.netcraft.com |

## 2.2 Key Capabilities

| Feature | Description | Use Case |
|---|---|---|
| **Phishing Detection** | Blocks known phishing and credential-harvesting pages in real time. | Stops victims from submitting credentials to cloned login pages. |
| **Site Report Panel** | Shows site rank, country, first seen date, and host from the browser toolbar. | Quick threat assessment without leaving the current page. |
| **Full Site Report** | Detailed network, hosting, IP delegation, DNS, and geolocation data. | Deep OSINT on a suspicious or attacker-controlled domain. |
| **Fraud Reporting** | Submit a malicious URL directly to Netcraft for investigation and takedown. | Report cloned pages or phishing sites for rapid removal. |
| **Malicious JS Blocking** | Detects and blocks harmful JavaScript injected into web pages. | Prevents drive-by downloads and browser-based exploit delivery. |
| **Netcraft Trends** | Historical site rank and hosting data for any domain. | Track how long a phishing domain has existed and who is hosting it. |

# 3. Extension Installation

## 3.1 Locating the Extension — Netcraft Website

The Netcraft browser extension is available directly from the official Netcraft website. Navigate to the Browser Extension page under Resources > Apps and Extensions. The page describes the extension's anti-phishing capabilities and provides download links for all supported browsers.

**Step 1:** Open a browser and navigate to: netcraft.com/resources/apps-and-extensions/browser-extension

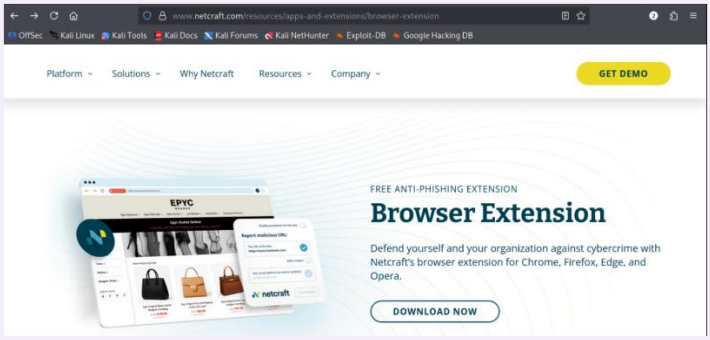



*Fig 3.1 — Netcraft website: Browser Extension page showing anti-phishing extension description and 'Download Now' button*

## 3.2 Choosing a Browser

The download page presents browser-specific icons for Google Chrome, Mozilla Firefox, and Microsoft Edge. Clicking the relevant icon redirects to that browser's official extension store — ensuring the extension is obtained from a verified, trusted source.

**Step 2:** Scroll to 'Download the Extension Today' section and click the Firefox icon to install on Firefox.

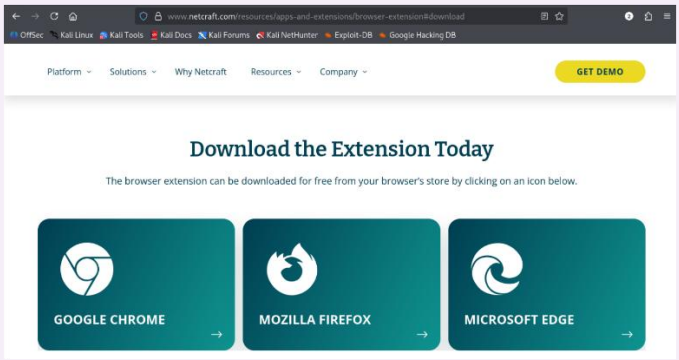



*Fig 3.2 — Netcraft: Download the Extension Today — Google Chrome, Mozilla Firefox, and Microsoft Edge options displayed*

## 3.3 Installing from the Firefox Add-Ons Store

Clicking the Firefox icon redirects to the Firefox Add-Ons store listing for the Netcraft Extension by Netcraft Ltd. The listing shows a 4.4 star rating and 3,298 active users. The extension provides comprehensive site information and protection from phishing and malicious JavaScript while browsing.

**Step 3:** On the Firefox Add-Ons page, click 'Add to Firefox' to install the Netcraft Extension.

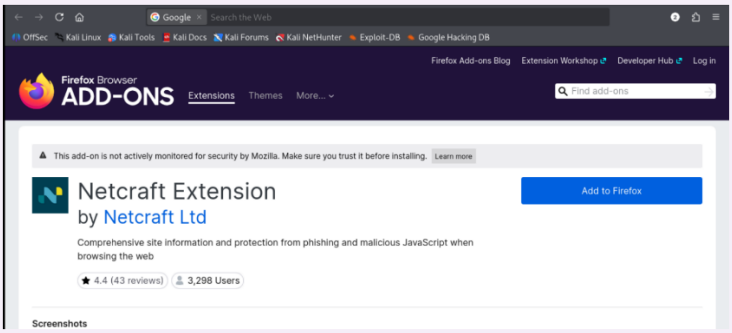



*Fig 3.3 — Firefox Add-Ons store: Netcraft Extension by Netcraft Ltd — 4.4 stars, 3,298 users, 'Add to Firefox' button*

**Note:** Firefox displays a warning that this add-on is not actively monitored for security by Mozilla. This is a standard notice for community extensions. Netcraft Ltd is a legitimate, established UK security company — the extension is safe to install.

## 3.4 Confirming Installation

After installation, the Netcraft Extension appears in the browser's Extensions panel in the toolbar. The extension icon is visible and active. The panel shows the extension name, its current permission status, and a settings gear icon for configuration.

**Step 4:** Click the Extensions icon in the Firefox toolbar to confirm the Netcraft Extension is installed and active.

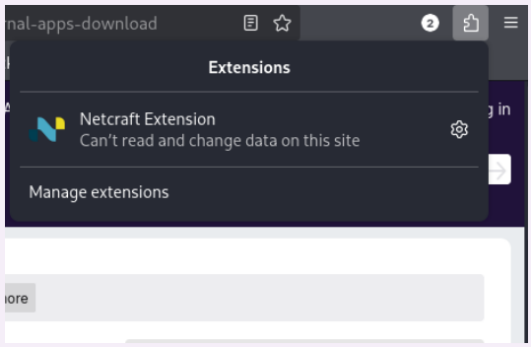



*Fig 3.4 — Firefox Extensions panel: Netcraft Extension listed — 'Can't read and change data on this site' status shown*

# 4. Using the Netcraft Extension

## 4.1 Live Site Report Panel

With the extension installed, clicking the Netcraft icon on any webpage opens an inline Site Report panel on the right side of the browser window. This panel shows real-time intelligence about the current website without navigating away from the page.

In this lab, the extension was tested on certifiedhacker.com to demonstrate its site inspection capabilities on a known security-related domain.

**Step 5:** Navigate to any website (e.g., https://certifiedhacker.com) and click the Netcraft extension icon in the toolbar.

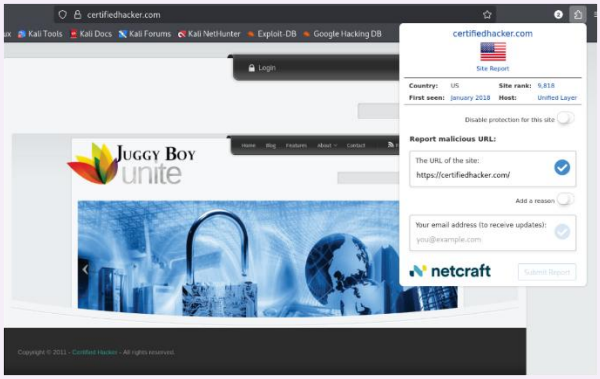



*Fig 4.1 — Netcraft extension active on certifiedhacker.com*

The Site Report panel displays the following information at a glance:

| | |
|---|---|
| **Country** | US — the country where the site's IP address is registered. |
| **Site Rank** | 9,818 — Netcraft's global popularity ranking based on web server surveys. |
| **First Seen** | January 2018 — when Netcraft first indexed this domain. |
| **Host** | Unified Layer — the hosting provider serving this domain. |
| **Report Malicious URL** | One-click form to submit the current URL to Netcraft's fraud team. |

**Note:** The 'First Seen' date is particularly useful for phishing detection — newly registered domains (days or weeks old) are a major red flag. Legitimate brands typically have domains dating back years. Cloned phishing pages hosted on fresh domains are easily flagged by this indicator.

### 4.2 Reporting a Malicious URL

The extension provides a built-in fraud reporting feature directly in the Site Report panel. The URL of the current page is pre-filled in the 'The URL of the site' field. An optional reason can be added, along with an email address to receive update notifications. Clicking Submit Report sends the URL to Netcraft's anti-phishing team for investigation and potential takedown.

## 5. Full Netcraft Site Report

Beyond the inline panel, Netcraft provides a full-page Site Report accessible at sitereport.netcraft.com. This report delivers comprehensive OSINT intelligence about any domain — covering background, network infrastructure, hosting, IP delegation, geolocation, and DNS history.

### 5.1 Accessing the Site Report

The full Site Report for certifiedhacker.com was accessed either by clicking 'Site Report' in the inline panel, or by directly visiting sitereport.netcraft.com and entering the domain in the lookup field.

**Step 6:** In the Netcraft extension panel, click 'Site Report' to open the full report page, or visit sitereport.netcraft.com and enter the domain.

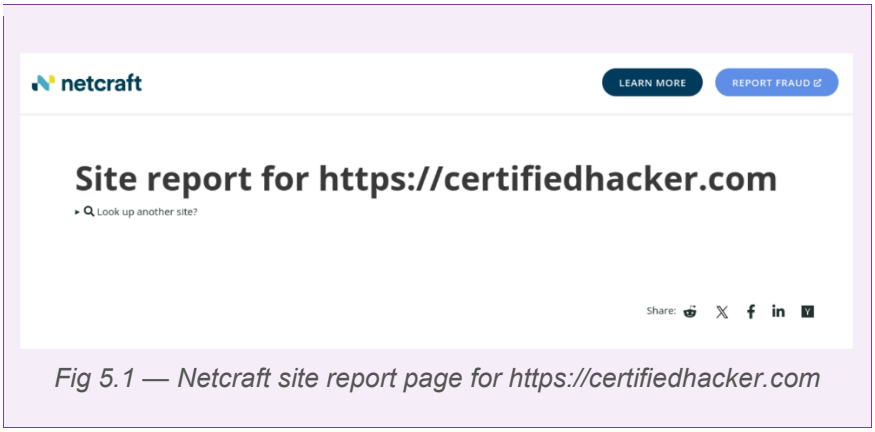



*Fig 5.1 — Netcraft site report page for https://certifiedhacker.com*

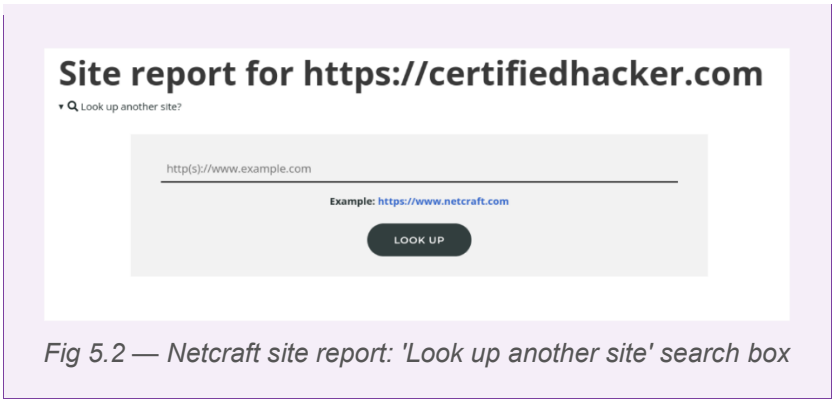



*Fig 5.2 — Netcraft site report: 'Look up another site' search box*

## 5.2 Background & Network Information

The Background section of the Site Report provides high-level details about the domain — its title, Netcraft site rank, primary language, and a brief description. The Network section reveals the hosting company, nameserver, domain registrar, IPv4/IPv6 addresses, and autonomous system (AS) number.

For the Netcraft.com report used as a reference in this lab, the following information was confirmed:

**Step 7:** Review the Background and Network sections of the full Site Report for hosting and infrastructure details.

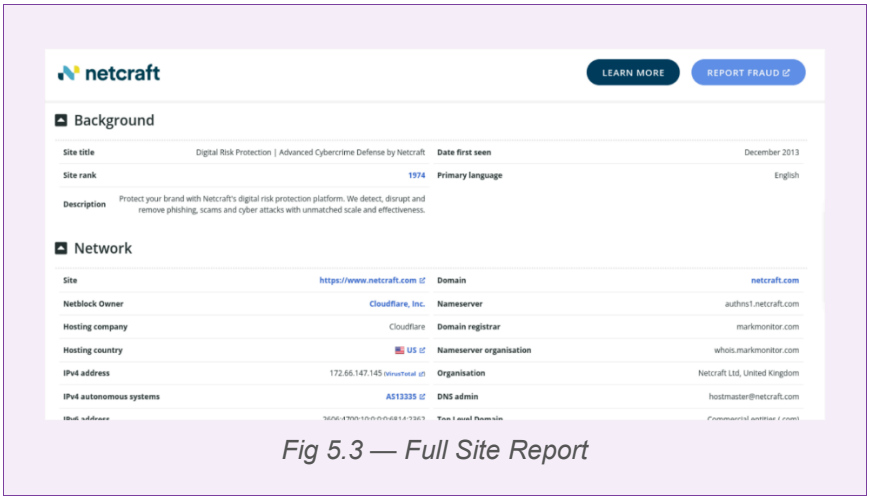



*Fig 5.3 — Full Site Report*

## 5.3 IP Delegation

The IP Delegation section traces the ownership hierarchy of the domain's IP address through IANA, regional internet registries (RIRs), and the final network block owner. This is useful for attribution — determining who ultimately controls the infrastructure hosting a suspicious site.

For the lab site (Netcraft.com at 172.66.147.145), the IP delegation chain confirmed that the address falls within Cloudflare's CLOUDFLARENET autonomous system — consistent with the hosting information shown in the Network section.

**Step 8:** Scroll to the IP Delegation section to trace the IP ownership chain from IANA to the final network block owner.

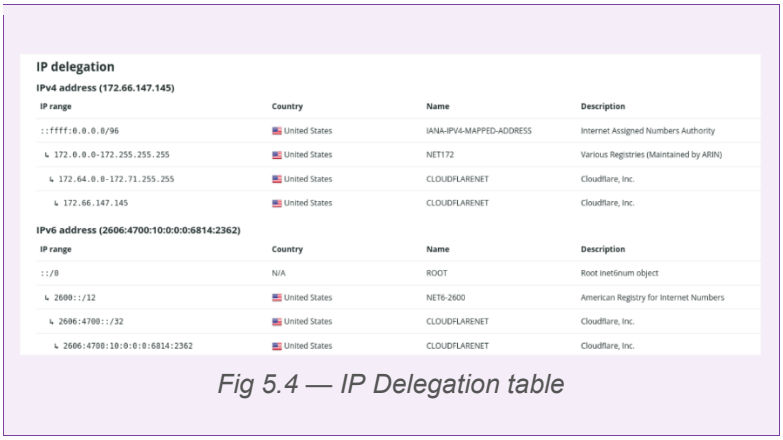



*Fig 5.4 — IP Delegation table*

## 5.4 IP Geolocation

Netcraft uses multilateration — measuring connection latency from multiple geographically distributed vantage points — to independently determine the physical location of a web server. This provides a more reliable geolocation than simple IP database lookups, which are often inaccurate for CDN and cloud-hosted sites.

The geolocation section displays an interactive world map highlighting the regions associated with the IP addresses serving the domain.

**Step 9:** Scroll to the IP Geolocation section to view the map and independently determined server location.




*Fig 5.5 — IP Geolocation*

# 6. Netcraft Trends

Netcraft Trends provides historical site ranking and hosting data, tracking how a domain's rank has changed over time and which hosting providers have served it. This is particularly useful for identifying newly established phishing domains or tracking infrastructure changes on known malicious sites.

## 6.1 Viewing the Trends List

The Netcraft Trends page lists the top-ranked sites globally, showing each site's URL, the date it was first seen by Netcraft, its hosting provider, and country flag. Sites can be looked up individually or browsed by rank range.

In the lab, the Trends page was accessed to show site rankings in the 9,800–9,814 range — contextualizing certifiedhacker.com's rank of 9,818 among similarly ranked global sites.

**Step 10:** Access the Netcraft Trends page from the Site Report or directly via the trends.netcraft.com URL.

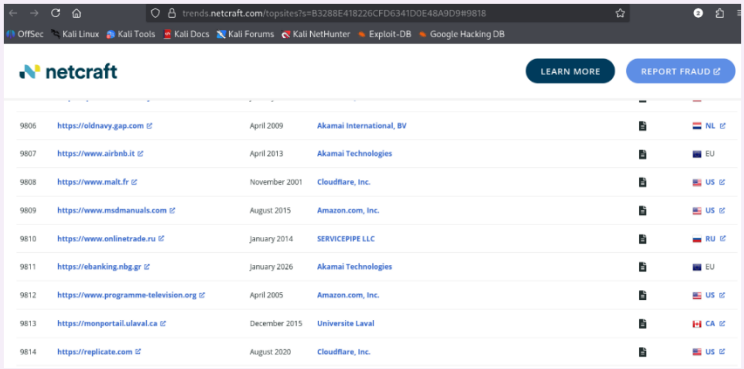



*Fig 6.1 — Netcraft Trends*

# 7. Findings Summary

| Component / Step | Status | Notes |
|---|---|---|
| **Extension Download (Netcraft site)** | ✓ Success | Extension download page located and browser options confirmed. |
| **Firefox Add-Ons Installation** | ✓ Success | Netcraft Extension installed successfully from Firefox Add-Ons store. |
| **Extension Active in Toolbar** | ✓ Success | Extension visible and active in Firefox Extensions panel. |
| **Inline Site Report Panel** | ✓ Success | Site Report panel displayed correctly on certifiedhacker.com — rank, host, country, first seen all confirmed. |
| **Full Site Report — Background** | ✓ Success | Hosting, registrar, IPv4, AS number, and language confirmed for Netcraft.com. |
| **IP Delegation Trace** | ✓ Success | IPv4 chain traced through IANA > NET172 > CLOUDFLARENET correctly. |
| **IP Geolocation Map** | ✓ Success | US-based server location confirmed via multilateration map display. |
| **Netcraft Trends Site Ranking** | ✓ Success | Trends page loaded, site rank 9,818 contextualized against similar-ranked global sites. |

## 8. Netcraft as a Defensive Countermeasure

In the context of this three-part lab series — SET, Gophish, and Netcraft — the Netcraft extension represents the defender's layer. The following table maps each attack technique from the SET/Gophish labs to how Netcraft detects or mitigates it:

| Attack Technique | How It Works (Attacker) | Netcraft Defense |
|---|---|---|
| **Credential Harvester (SET)** | Clones a real login page; victim submits credentials to attacker's server. | Flags new/unknown domain, shows 'First Seen' recently — warns before credentials are entered. |
| **Site Cloner + Gmail Link Spoof (SET)** | Email link displays legitimate URL but redirects to attacker's cloned page at a local/unknown IP. | Site Report shows IP with no history or unknown host — user can verify before submitting data. |
| **Gophish Phishing Campaign** | Sends branded phishing email; landing page harvests credentials and redirects to legitimate site. | Known phishing URLs blocked outright. Users can submit landing page URL to Netcraft fraud team for rapid takedown. |

## 9. Conclusion

This lab successfully demonstrated the installation and use of the Netcraft Anti-Phishing Browser Extension as a real-world defensive tool. All 11 screenshots covering the extension download, installation, inline site inspection, full site report, IP delegation, geolocation, and Netcraft Trends were documented and labeled.

Netcraft provides a practical, zero-cost layer of browser-level protection that directly counters the phishing techniques demonstrated in the SET and Gophish labs. Its Site Report data — particularly First Seen date, hosting provider, and IP delegation — gives users and analysts the contextual information needed to identify suspicious pages before credentials are submitted.